MadePlus SHOEML – Training Tab: Improvement Plan

Overview
The current Training tab:
• Records one labeled session at a time (single label for whole run).
• Windows it (3 s, hop 1.5 s).
• Computes simple stats (mean/var/RMS of roll, pitch, yaw, cadence, stride, temp).
• Trains a softmax classifier only on that buffer.
• Shows one accuracy number and saves the model JSON.

This document lists staged upgrades to make training richer, more robust, and more useful.

Layer 1 – Better Data & UI

1. Train on all saved sessions
• New button: “Train on all data”.
• Read all CSVs from shoeml/sessions.
• Parse label from filename and label column.
• Build one combined dataset of windows across all sessions.
• Split 80/20 train/validation on this combined dataset.
Benefit: Model learns from many walks/runs instead of only the latest recording.

2. Dataset summary card
• On Training tab, show a small card summarizing counts per label, e.g.:
  – walking: 540 windows
  – running: 220 windows
  – stairs_up: 80
  – stairs_down: 65
  – standing: 150
  – sitting: 130
Benefit: Quickly see imbalanced data before training.

3. Session metadata
• After recording stops, show a short dialog:
  – surface (road / treadmill / grass / stairs)
  – foot (left / right)
  – notes (free text)
• Store as # meta lines at top of CSV and as properties in GeoJSON.
Benefit: Later filtering and training “outdoor-only” or “treadmill-only” models.

Layer 2 – Better Training & Evaluation

4. Train/validation statistics
• In addition to a single accuracy, compute:
  – train accuracy
  – validation accuracy
  – number of windows per label for each split
• Show these in:
  – SnackBar
  – Training summary dialog
  – Stored in the model JSON as metadata (train_acc, val_acc, num_windows).
Benefit: See if model is overfitting or if validation data is too small.

5. Confusion matrix per label
• On the validation set, build a confusion matrix:

– C[true_label][pred_label] = count of windows.
• Show as a simple grid/table for each label.
Benefit: See which activities are confused (e.g. standing vs sitting).

6. Model management
• New “Models” card on Training tab:
– List saved models from shoeml/models.
– Show: name, date, train acc, val acc, labels.
– Allow selecting one as “Active model”.
– Add delete icon to remove old models.
• Store metrics inside JSON alongside W, b and labels.
Benefit: Treat models as first-class objects and keep the good ones.

Layer 3 – Live Inference / Test Mode

7. Live prediction mode
• New section (either new tab or card on Dashboard):
– Uses currently active model.
– Builds a sliding window from recent _RawSample data.
– Runs SoftmaxClassifier.predict(features) every second.
– Displays:
· Current predicted label (walking / running / stairs / standing / sitting).
· Confidence (max softmax probability).
Benefit: Immediate feedback that training worked and quick spotting of bad predictions.

8. Record test sessions with predictions
• Option in Live mode:
– Toggle “log predictions”.
– Save CSV with time, features, predicted label and confidence.
– Optionally, big button to mark ground truth label during test.
Benefit: Gather more evaluation data and new labels with minimal effort.

Layer 4 – Smarter Features

9. Extended feature set inside _features()
Time-domain dynamics:
• Mean absolute derivative of roll, pitch, yaw.
• Peak-to-peak value (max - min).
• Zero-crossing rate (how often signal crosses mean).

Frequency-domain ideas (if you add raw accel/gyro):
• FFT magnitude of vertical acceleration.
• Dominant frequency (step rate).
• Energy in specific frequency bands.

Gait-specific:
• Average step count per second in window.
• Variance of stride length.
• Ratio of uphill vs downhill steps using delta-h.

Benefit: Richer features improve separability between activities.

Layer 5 – UX polish for Training

10. Minimum session length
• Before enabling training from a session, enforce:

– Duration >= e.g. 10–15 seconds.
– OR distance >= e.g. 20–30 meters.
Benefit: Avoid noisy models from tiny accidental recordings.

11. Calibration wizard
• Guided flow for new users:
  – Step 1: “Record 30 s walking”.
  – Step 2: “Record 30 s running”.
  – Step 3: “Record stairs up/down”.
  – Automatically trains a “Quick Calibrated Model” at the end.
Benefit: Easy on-boarding and consistent baseline model for each shoe/user.

12. Dataset export
• Button to “Export dataset”:
  – Zips selected CSV/GeoJSON/GPX/KML files.
  – Shares via the standard Android share sheet.
Benefit: Easy offline analysis in Python notebooks or desktop tools.

Summary
Start with:
• Train on all sessions.
• Dataset summary.
• Better train/validation metrics and confusion matrix.
Then move toward live prediction mode and richer features. This staged approach lets you keep the app stable while steadily making the Training tab more powerful for real-world experiments.